

learning enters here

supplied by the analysis, not the box

Game Γ ,
deviation class $\mathcal{D}$

computed
quantity

candidate $\hat{x}$

guarantee
no profitable
deviation over $\mathcal{D}$

*what learning
replaces*

*source of the
guarantee*

target it certifies

① **Convergence
proof**

regularized
response map

a convergence
theorem

regularized Nash
(two-player zero-sum)

② **Regret**

the regret /
value estimate
(CFR, Deep CFR)

regret
decomposition
+ duality

Nash
(CCE / CE in
general-sum)

③ **Self-play**

the whole policy
(AlphaZero)

none — empirical
strength only

no equilibrium
certified